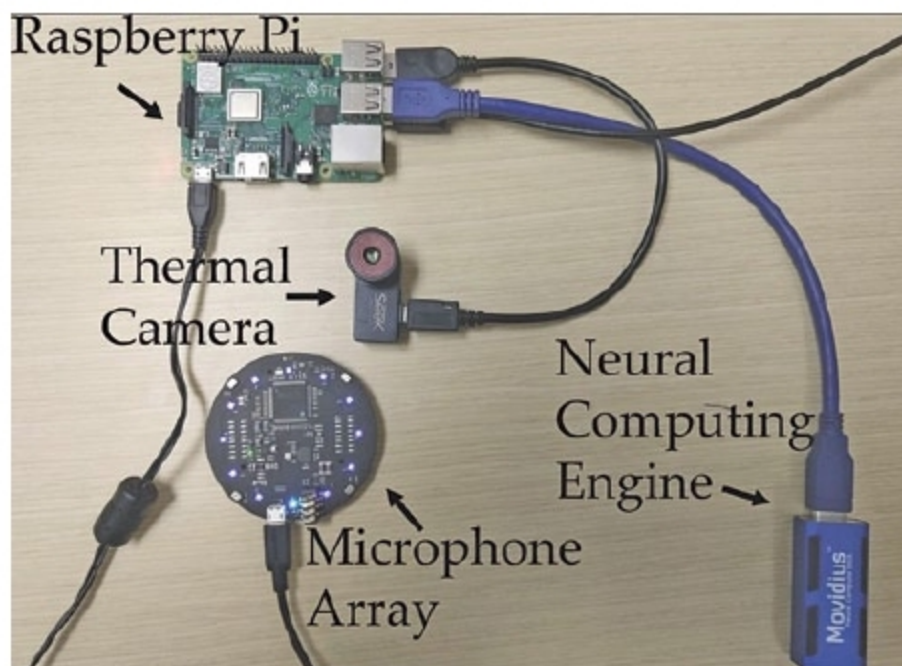




Device to predict flu pandemics in advance developed

A portable surveillance device that can detect coughing and crowd size in real time, analyse the data to directly monitor flu-like illnesses and influenza trends, and predict the next



The FluSense device houses these components

pandemic in the making has been invented. Called FluSense, and powered by machine learning, it has been developed by a team of US researchers from the University of Massachusetts. It is likely to find use in hospitals, healthcare waiting rooms and larger public gatherings as a surveillance tool to forecast seasonal flu and other viral respiratory outbreaks, such as Covid-19 novel coronavirus pandemic. According to study co-author Tauhidur Rahman, assistant professor of computer and information sciences, "This may allow us to predict flu trends in a much more accurate manner."

The FluSense platform processes a low-cost microphone array and thermal imaging data with a Raspberry Pi and neural computing engine. The researchers first developed a lab-based cough model. They then trained the deep neural network classifier to draw bounding boxes on thermal images representing people, and then to count them. From December 2018 to July 2019, the FluSense platform collected and analysed more than 350,000 thermal images and 21 million non-speech audio samples from the public waiting areas. FluSense was able to accurately predict daily illness rates at the university clinic. The next step is to test it in other public areas and geographic locations.

Study says 5G is safe to use

There are rumours going on in social media that radiation from radio waves used for various communication modes are causing pandemics like novel coronavirus. But The International Commission on Non-Ionizing Radiation Protection (ICNIRP) has quashed these rumours and said there is no evidence showing 5G mobile networks pose any threat to human health. ICNIRP has spent seven years examining the scientific evidence and has updated its guidelines to improve protection for 5G technologies that use high frequencies.

ICNIRP said the 1998 guidelines on protecting humans from radiation from phone networks, Wi-Fi and Bluetooth would have been appropriate for 5G technology, but frequencies over 6GHz may need more conservative standards.

"The guidelines have been developed after a thorough review of all relevant scientific literature, scientific workshops and an extensive public consultation process. They provide protection against all scientifically substantiated adverse health effects due to EMF exposure in the 100kHz to 300GHz range," said ICNIRP chairman Dr Eric van Rongen.

The ICNIRP suggested restrictions that would stop whole-body exposure to the 6GHz frequency and intense exposure to small areas of the body, though this is not mandatory. "The most important thing for people to remember is that 5G technologies will not be able to cause harm when these new guidelines are adhered to," van Rongen added. Mobile trade group GSMA said current 5G phones already fall within the new standards' limits.

Work from home culture likely to grow for tech workers

While the outbreak of novel coronavirus is forcing people to work from home in India, it might become a norm for a majority of office workers. It is estimated that more than half of the approximately three million Indian IT workers are now

working from home thanks to social distancing imposed by the outbreak. If they continue to work from home for a few more months, at least IT companies are likely to conclude that nearly one-third of their workforce can operate remotely.